

Olivia Liao

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EDUCATION

University of Southern California (Expected Graduation: May 2027)

Master's in Applied Data Science, Bachelor's in Computer Science (GPA: 3.56/4.0)

Los Angeles, CA

Fall 2024 – Present

EXPERIENCE

AI Research Intern

Jun 2026 – Aug 2026

Tanium

- Diagnosed agent memory retrieval and context persistence failures using Claude for agent code and data handler instrumentation to emit telemetry; queried 23K+ sessions across 100K+ endpoints via GraphQL
- Discovered 21% verdict mislabeling after auditing LLM evaluations; redesigned label taxonomy and reworked agent's system prompt to relabel and additionally capture 13% more negative human feedback events
- Achieved 71% AUC in feasibility testing of latency-aware, turn-level risk prediction using frozen encoder, sliding context window, cumulative-history feature, and lightweight gradient-boosted (XGBoost) heads

Data Science Research Intern

May 2024 – Feb 2026

Worcester Polytechnic Institute | [Publication](#)

Worcester, MA

- First-authored publication with cross-functional team, presented peer-reviewed work at *BIOINFORMATICS 2026*
- Introduced novel AutoEncoder-CNN architecture processing 20K+ genes from 146 entities, delivering biologically-significant timing predictions within 1-hour error and demonstrating cross-region generalization
- Validated 68% improvement over baseline through systematic hyperparameter tuning and cross-validated model selection across 700+ configurations, including 4 dimensionality-reduction methods and 16 regression models

Business Analytics Extern

Sept 2025 – Jan 2026

Amazon, Extern | Operational Strategy & People Analytics

Remote

- Built RAG-based GPT-4o-mini classifier with LangChain orchestration and embedding-based semantic retrieval; ran 30 prompt engineering A/B experiments, achieving 81% classification accuracy and 90% retrieval precision
- Corrected evaluation leakage (94% to 91.9% F1) and selected final model via LLM-as-judge evaluation guardrails (confidence thresholds, hallucination analysis); delivered KPI insights to cross-functional stakeholders

Machine Learning Research Intern

May 2025 – Aug 2025

University of Wyoming

Laramie, WY

- Achieved 89% accuracy on multimodal time-series data and 72% accuracy on medical imaging for Parkinson's Disease detection by implementing ResNet18 CNN with transfer learning
- Overcame limited sample size by applying SMOTE data augmentation and hybrid PCA-HOG feature extraction pipeline; enhanced model generalization by 18% and reduced overfitting by 23%

Machine Learning Research Intern

Aug 2024 – Jan 2025

USC Informatics & Computing in Neuroscience Lab

Los Angeles, CA

- Developed signal-processing XGBoost model achieving 71% cross-validation accuracy by engineering 108 multimodal features from EEG recordings and clinical assessments to predict treatment outcomes
- Designed SHAP explainability framework identifying 10 key neurological biomarkers ranked by predictive importance, enabling individualized predictions to build physician trust for regulatory approval

PROJECTS

RetailPRED: End-to-End ML Forecasting System

Oct 2025 – May 2026

- Built and deployed MLOps pipeline on GCP using Cursor and Codex for prototyping and testing; enabled weekly model forecasts, validation, and performance monitoring with 90.9% historical accuracy
- Achieved 10× reduction in forecast error on Random Forest-LightGBM ensemble through ablation of 73 engineered time-series features; surfaced 2,214 validated forecasts through React and Tableau

TECHNICAL SKILLS

Languages & AI-Assisted Workflows: Python (PyTorch, TensorFlow, scikit-learn, Pandas), SQL, C/C++, Claude, Codex

Agentic AI: LLM Evaluation, Prompt Engineering, Human-Feedback Analysis, Autoencoders, Computer Vision, Time-Series Forecasting, Anomaly Detection, Dimensionality Reduction, Hyperparameter Tuning, Model Explainability

ML Engineering: XGBoost, LightGBM, Random Forest, Deep Learning, CNNs, ML Pipelines, Feature Engineering, Feature Selection, Model Evaluation, Model Monitoring, Inference, Airflow, GCP, FastAPI, REST APIs, Docker, Git, CI/CD